CYBER-PHYSICAL SECURITY · RESEARCH SHOWCASE

The lie the network can't see.

A digital-twin and physical-watermarking testbed for multi-layer intrusion detection in SCADA systems.

S. Saurabh Krishnan · Hemant Bhupesh · Sanjai S · Nitesh V. M.

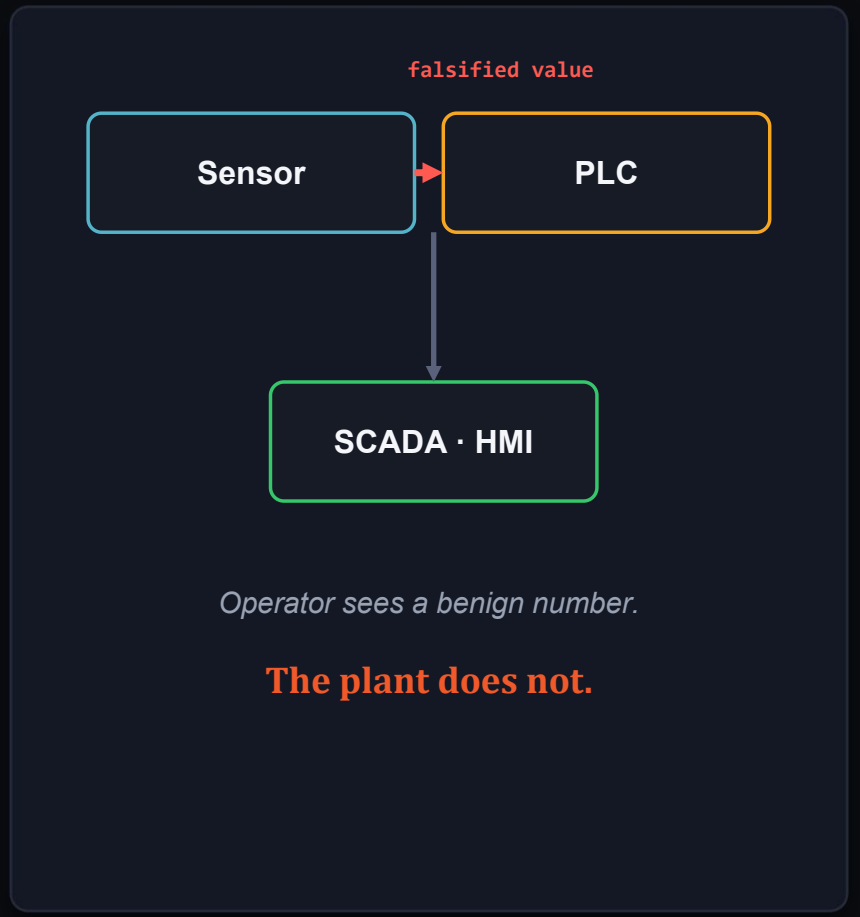
Amrita Vishwa Vidyapeetham · Coimbatore, India

THE BLIND SPOT

A compromised sensor sends a perfect packet with a false value.

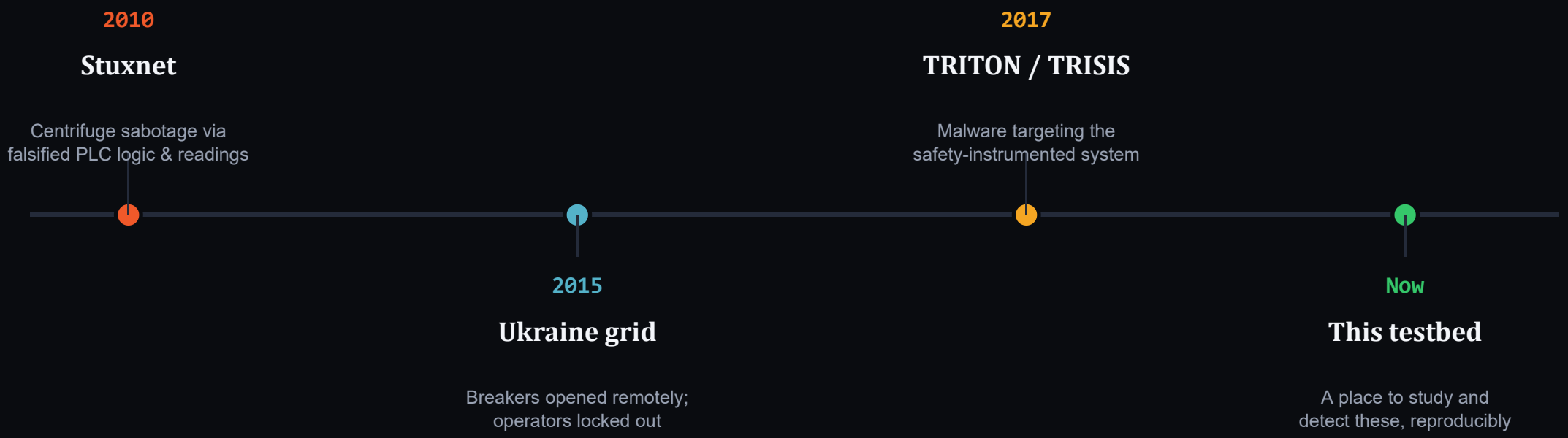
The network layer sees nothing wrong — the frame is well-formed and on time. **Meanwhile the real process diverges.**

Detecting this needs reasoning about the physics of the process, not just its packets.



IT HAS ALREADY HAPPENED

Fifteen years of attacks that reached the physical layer.



Dates and incidents are historical fact; the testbed is a simulation.

THE TESTBED

A liquid-metal boiler plant, alive at 10 Hz.



Operator HMI — animated P&ID, live telemetry, independent safety panel.

37
sensors

17
actuators

5
PLCs / RTU

72
network nodes

10 Hz
live engine

56
tests passing

THE SIMULATED NETWORK

Purdue-zoned, protocol-a

Modbus TCP as spec-conformant ADU bytes; **OPC UA** and **MQTT** emulated. A discrete-event bus adds latency, loss and MITM, and mirrors supervisory traffic to the IDS through a SPAN port.



Live SPAN-mirrored Modbus / MQTT traffic feeding the four-layer IDS.

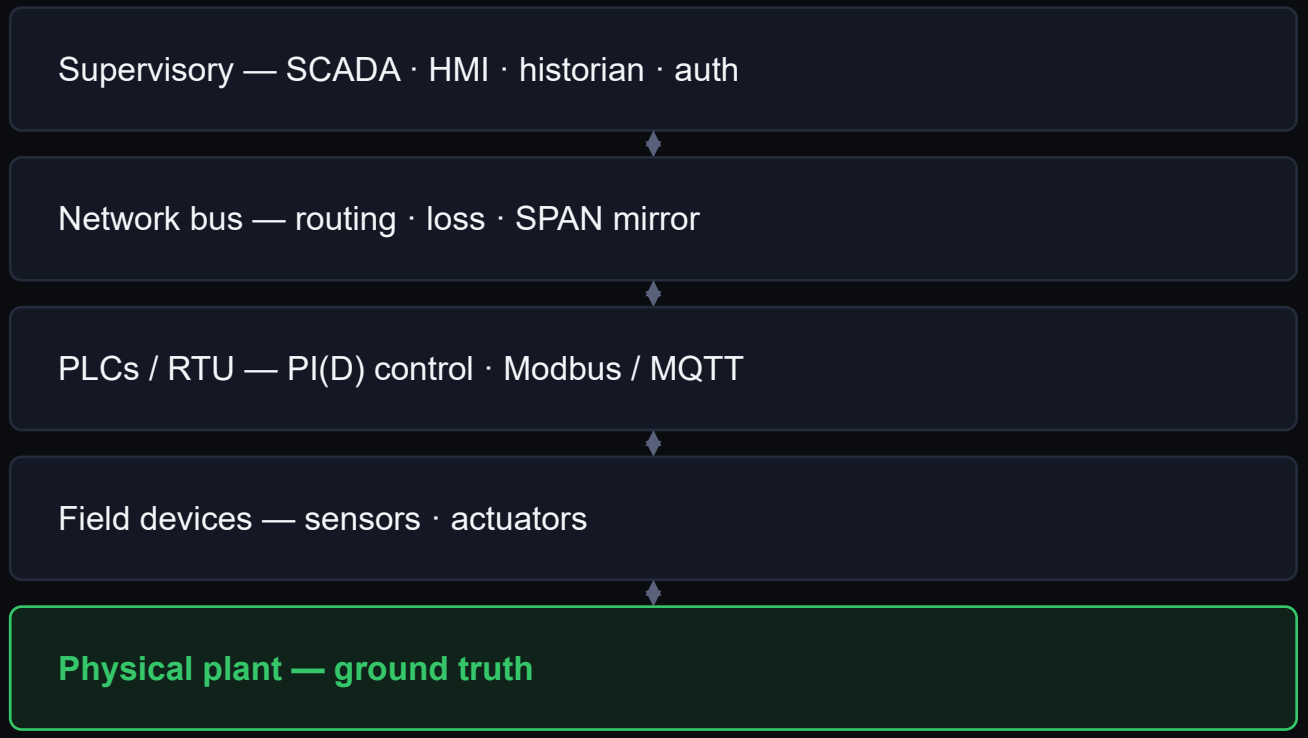
01

THE SYSTEM

How the plant, the twin, the detectors and the watermark fit together.

ARCHITECTURE

Five layers, watched by four observers.



OBSERVERS

- **Digital twin** — independent physics + residuals
- **Four-layer IDS** — network · process · twin · watermark
- **Correlation** — dedup · classify · fuse confidence
- **Response** — monitor / approve / automatic

Invariant: the attacker touches only the network and field devices — never the twin, IDS, watermark or safety system.

THE 10 HZ LOOP

One tick, in a fixed order.

1. Attack lab applies manipulations
2. Physical plant integrates (ground truth)
3. Hard-wired safety (SIS) may trip
4. Field I/O — sensors sample, PLCs scan
5. Network bus delivers (SPAN mirror)
6. SCADA polling · historian · twin feeds
7. Digital twin + watermark update
8. IDS (4 layers) → correlation
9. Response engine → snapshot

What they never see

Green steps read the **true plant state.**

The attacker (step 1) acts only through network and field hooks.

The IDS (steps 5–8) sees only the SPAN mirror — the same bytes a real tap would capture.

THE INDEPENDENT TWIN

A model that won't absorb a lie.

The twin runs its **own RK4 physics** with datasheet parameters — it never reads the plant's hidden state. Each tick it predicts, then fuses each measurement through a **gate**.

A reading that jumps outside the gate (|z| < 5) is rejected — so a stealthy lie shows up as a growing residual instead of quietly corrupting the estimate.

```
r  =  observed  -  predicted

z  =  ( r - bias ) / σ
```

IN THE LIVE FALSE-DATA DEMO

real 459.8 °C

vs falsified 442.3 °C shown to the operator

twin residual z = −24.9

FOUR LAYERS, ONE VERDICT

Each layer covers another's blind spot.

NETWORK

rogue devices · spoofing · floods · replay · unauthorised writes

PROCESS

limits · frozen signals · rate-of-change · setpoint deviation · energy balance

TWIN

persistent & sudden residuals · correlated multi-sensor anomalies

WATERMARK

missing or abnormal response to the injected excitation

Correlation engine: deduplicates findings into one incident, fuses confidence $C = 1 - \Pi(1 - c_i)$, raises severity per corroborating layer, and classifies.

PHYSICAL WATERMARKING

Probe the process, then listen for the echo.

1 INJECT

A bounded PRBS excitation is added to the pump command inside the PLC's secured block. The exported register still reports the base value.

2 PREDICT

Two shadow copies of the twin's hydraulics compute the expected response to the excitation.

3 COMPARE

Observed flow & pressure are matched to the expected response by correlation ρ, gain and SNR.

Decision: consistent / suspicious / inconclusive Self-suspends on a safety trip, saturation or poor measurement quality — never forces an unsafe perturbation.

02

THE EVIDENCE

Ten attacks, one honest question: which layers actually catch them?

THE ATTACK LABORATORY

Ten reproducible scenarios.

1. False sensor-data injection
2. Frozen sensor readings
3. Actuator-response manipulation
4. Unauthorised setpoint write
5. Command replay
6. Supervisory-link DoS
7. Insider / out-of-policy write
8. Digital-twin model tamper
9. Watermark interference
10. Coordinated process + network

Each records its seed, makes real changes, exposes a ground-truth label, and supports stop/recover. Detection must come from evidence — starting a scenario raises no alert by itself.

LIVE — FALSE DATA INJECTION

The twin tells the truth the sensor hides.



459.8 °C

real hot-leg temperature

442.3 °C

shown to the operator

z = −24.9

twin residual, CUSUM saturated

96 %

incident confidence

Observed (blue) is pulled to the fake value; the twin prediction (orange) keeps tracking reality.

CORRELATED, THEN CONTAINED

Two layers fire. One incident. An automatic response.

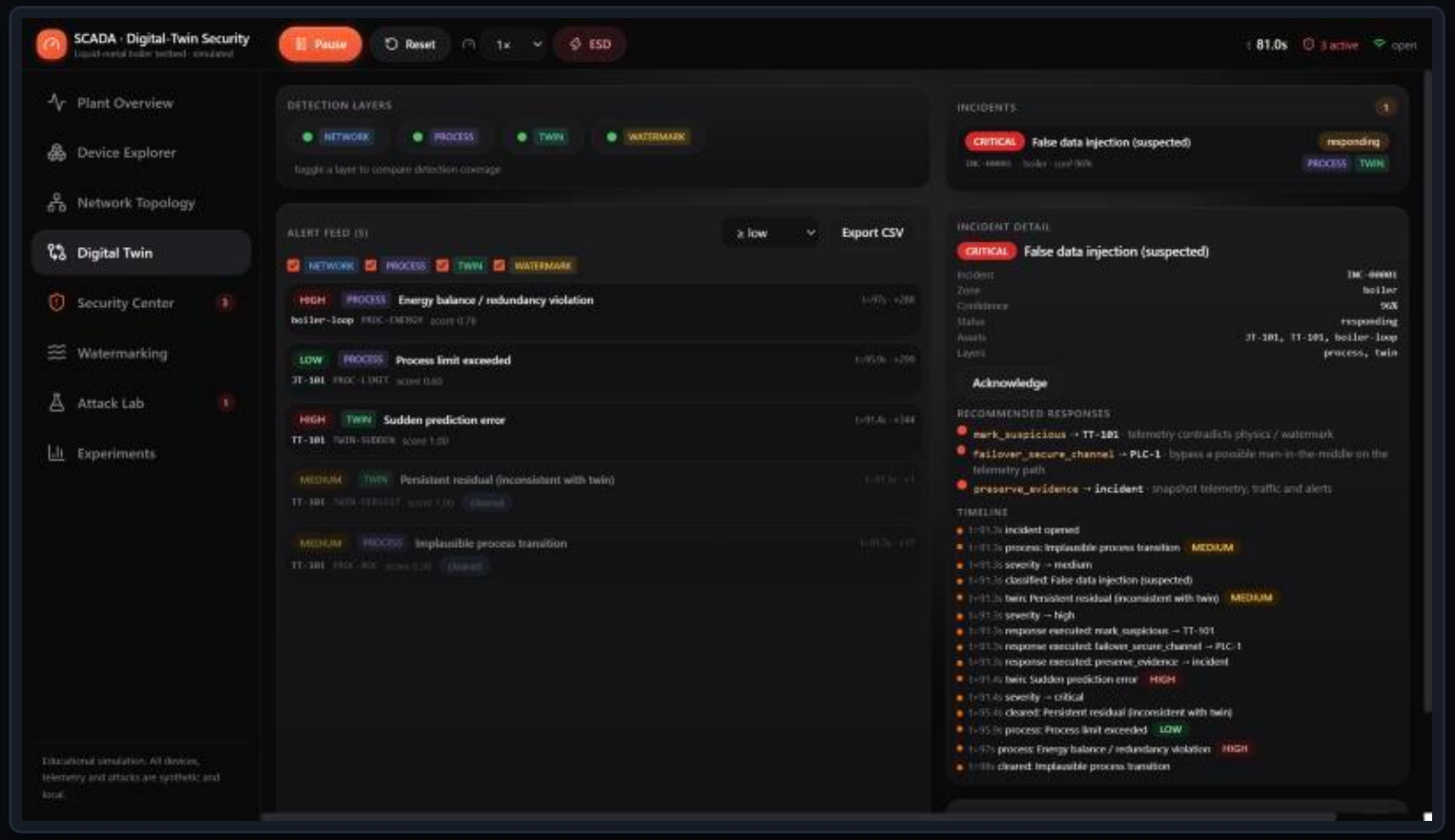
PROCESS + TWIN fire
energy-balance + sudden-residual evidence — the network stays silent

↓

Correlated incident
“False data injection” · critical · 96% confidence

↓

Automatic response
mark suspicious · secure-channel failover · preserve evidence



Alert feed, the correlated critical incident, and the executed responses.

RESULTS

Detection F1 by configuration.

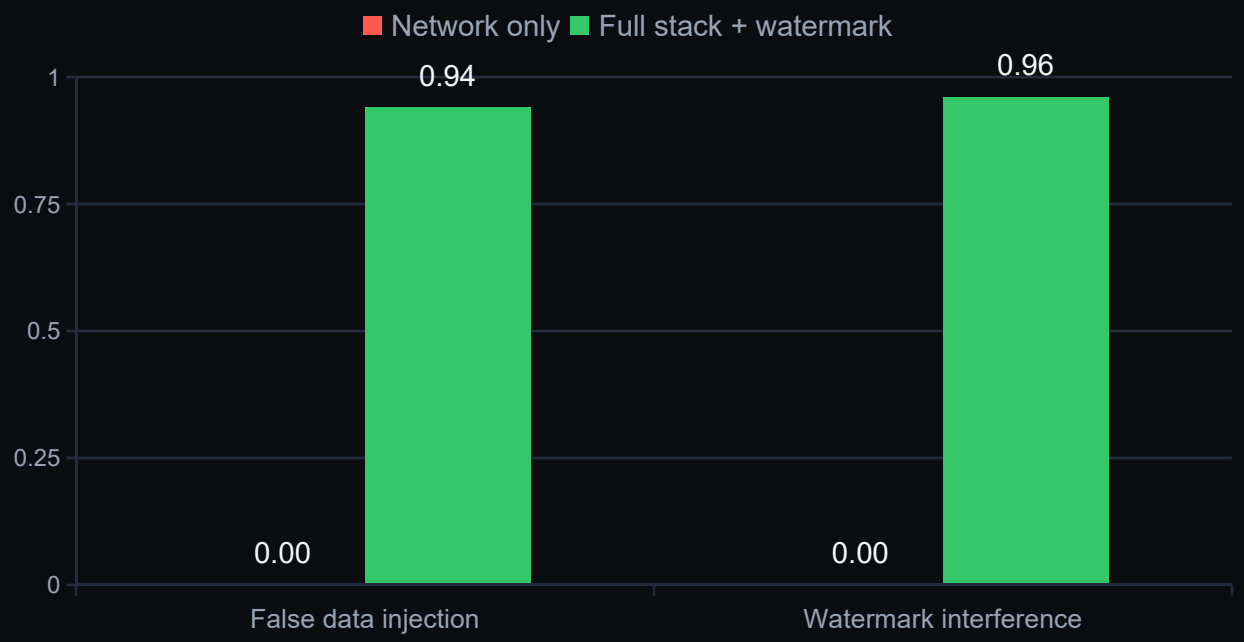
Scenario	Net only	Net+Proc	+Twin	Full+WM	Latency
A. Normal operation	—	—	—	—	—
B. False data injection	miss	0.94	0.94	0.94	0.2 s
C. Watermark interference	miss	miss	0.71	0.96	1.1 s
D. Unauthorised setpoint	0.91	0.91	0.91	0.91	0.0 s
E. Command replay	0.93	0.93	0.93	0.93	0.5 s
F. Coordinated	0.91	0.91	0.91	0.91	0.5 s

Network-only is blind to B and C — field-bus lies leave no network evidence. Process + twin recover B in 0.2 s; the watermark lifts C from 0.71 to 0.96.

THE KEY FINDING

Zero, or ninety-four.

For telemetry-falsification attacks, a packet-only IDS scores nothing. The extra layers do all the work.



THE EVIDENCE

THE WHOLE ARGUMENT

Command-path attacks

caught by the network layer.

Telemetry falsification

invisible there — caught by process + twin.

Erased signatures

caught by the active watermark.

WHO BUILT WHAT

Author contributions.

S. Saurabh Krishnan

Digital twin — model, gated correction, residual/CUSUM analysis — and the experiment & evaluation framework and 56-test suite.

Hemant Bhupesh

Plant & network simulation — boiler and subsystems, device abstractions, PLC/RTU control, the safety system and the protocol layer.

Sanjai S

Multi-layer IDS and correlation, the ten attack scenarios, and the automated defensive-response engine.

Nitesh V. M.

Physical watermarking, the React operator HMI, the FastAPI/WebSocket backend and persistence, and documentation.

DEFENCE IN THE PHYSICS

A network-only defender is blind.
A twin, the physics and a watermark can see.

Full multi-layer stack: sensor falsification detected in 0.2 s (F1 0.94) where network-only scores 0.00. Live at 10 Hz, reproducible, 56 tests.

Thank you.

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Amrita Vishwa Vidyapeetham, Coimbatore · educational research simulation

Refs: Stuxnet (Symantec, 2011) · Mo & Sinopoli (Allerton 2009) · Mo et al. (IEEE CSM 2015) · Cárdenas et al. (ASIACCS 2011) · Urbina et al. (CCS 2016) · Goh et al. (CRITIS 2016) · Lee et al. (E-ISAC/SANS 2016).